

Question 1 (15 Marks)**Marks**

- (a) Draw a neat sketch of $xy = 8$, **clearly** indicating on the sketch, the coordinates of the foci, vertices, and the equations of the directrices. **4**

- (b) A raindrop falls so that its velocity v m/s at time t seconds is given by

$$\frac{dv}{dt} = \frac{1}{3}(3g - 2v)$$

where g is the acceleration due to gravity.

- (i) Show that $v = \frac{3g}{2} \left(1 - e^{-\frac{2}{3}t} \right)$. **3**

- (ii) Find the limiting velocity of the raindrop in terms of g . **1**

- (iii) Find the time when the velocity reaches $\frac{1}{2}g$ m/s. **2**

- (c) The rate of increase of the population, $P(t)$, of a particular bird species at time t years is given by the equation:

$$\frac{dP}{dt} = kP(Q - P)$$

where k and Q are positive constants and $P(0) < Q$.

- (i) Verify that the expression $P(t) = \frac{QC}{C + e^{-kQt}}$, where C is a constant, **3**

is a solution of the equation.

- (ii) Describe the behaviour of P as $t \rightarrow \infty$. **1**

- (iii) Describe what happens to the rate of increase of the population as $t \rightarrow \infty$. **1**

Question 2 (15 Marks) (Start a new page)

- (a) A particle of unit mass is projected vertically upwards from the ground with initial speed u m/s. If air resistance at any time t seconds is proportional to the velocity at that instant, and assuming air resistance is $-kv$,

- (i) Prove that if the highest point is reached by the particle in time T seconds then **4**

$$kT = \log \left(1 + \frac{ku}{g} \right)$$

where g is the acceleration due to gravity.

- (ii) If the highest point reached is at a height h metres above the ground, **5**
prove that $hk = u - gT$.

Question 2 continues on the next page

Question 2 continued**Marks**

- (b) The normal at a variable point $P\left(2p, \frac{2}{p}\right)$ on $xy = 4$, given by $y = p^2x - 2p^3 + \frac{2}{p}$, meets the x – axis at Q .
- (i) Find the coordinates of Q . 1
- (ii) Find the coordinates of the midpoint, M , of PQ . 2
- (iii) Hence, find the locus of M . 3

Question 3 (15 Marks)

- (a) (i) Prove that the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab square units. 3
- (ii) Hence, by the **method of cylindrical shells**, find the volume of the solid formed when the area is rotated through 1 complete revolution about the line $y = b$. 3
- (b) The area enclosed by the graph of the function $y = e^{2x}$, the y – axis and the horizontal line $y = e^2$ is rotated about the y – axis.
- (i) Show that the volume is given by $\Delta V = \sum_0^1 2\pi x(e^2 - e^{2x})\Delta x$. 2
- (ii) Hence, find the exact volume of the solid of revolution formed. 3
- (c) If the gradients of the tangents drawn to the curves $xy = c^2$ and $y^2 = 4ax$ at the point of intersection are m and M respectively. Prove that $m = -2M$. 4

Question 4 (15 Marks)**(Start a new page)****Marks**

- (c) A particle moves in a straight line so that it's acceleration is inversely proportional to the square of its distance from a point O in the line and is directed towards O . It starts from rest at a distance a units from O .
- (i) What is its velocity when it first reaches a distance, $\frac{a}{2}$ units, from O ? 3
- (ii) Show that the time taken to first reach this distance in part (i) is given by 3

$$t = \frac{(\pi + 2)a^{\frac{3}{2}}}{4\sqrt{2k}}, \text{ where } k \text{ is a constant,}$$

$$\text{given that } \frac{d}{dx} \left[\sqrt{x(a-x)} + \frac{a}{2} \sin^{-1} \left(\frac{a-2x}{a} \right) \right] = -\sqrt{\frac{x}{a-x}}$$

Question 4 is on the next page.

Question 4 continued**Marks**

- (b) A is the area of the region R bounded by the upper branch of the hyperbola

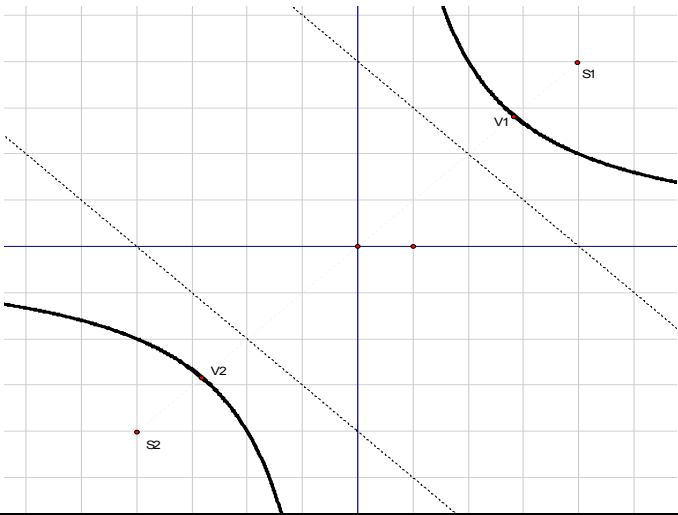
$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1, \text{ the } x\text{-axis and the lines } x = \pm a.$$

- (i) Show that $A = \frac{Lb}{2} [\sqrt{2} + \ln(1 + \sqrt{2})]$ square units, where L the length of the base of R is $2a$ units. **5**

- (ii) S is the solid whose base is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Cross sections perpendicular to the base and to the minor axis, are plane figures similar to region R where the line of intersection of the planes is the base length of R . Find the volume of S . **4**

~ END OF TEST ~

2006 JRAHS Extension 2 Term 2 Assessment / LK

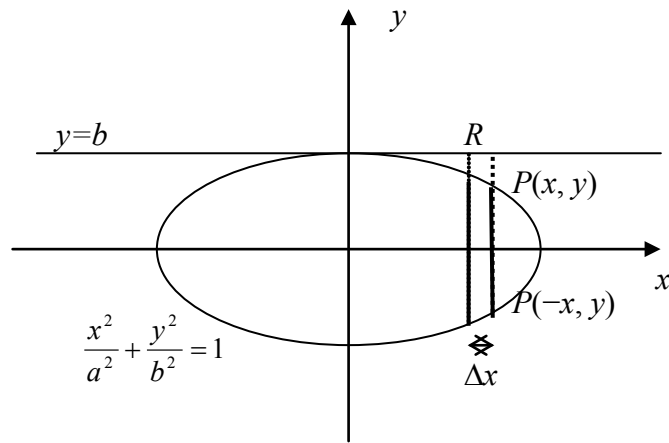
Solutions to Questions	Marking Scheme	Comments
Question 1		
(a) 	- Foci $(\pm 4, \pm 4) \rightarrow 1 \text{ mk}$ - Vertices $(\pm 2\sqrt{2}, \pm 2\sqrt{2}) \rightarrow 1 \text{ mk}$ - Eqn. of directrices $x + y = \pm 4 \rightarrow 1 \text{ mk}$ - Shape of graph $\rightarrow 1 \text{ mk}$	Students must clearly show the coordinates of the foci and vertices & eqn. of directrices to obtain full marks. ½ mark off is scale is wrong/ no scale given.
(b) (i) $\frac{dv}{dt} = \frac{1}{3}(3g - 2v)$ $\therefore \int_0^v \frac{dv}{3g - 2v} = \frac{1}{3} \int_0^t dt$ $\therefore -\frac{1}{2} \ln 3g - 2v _0^v = \frac{1}{3} [t]_0^t$ $\therefore -\frac{1}{2} \ln \left \frac{3g - 2v}{3g} \right = \frac{1}{3} t \quad \text{or} \quad \frac{1}{2} \ln \left \frac{2v - 3g}{3g} \right = \frac{1}{3} t$ $\therefore \ln \left \frac{3g - 2v}{3g} \right = -\frac{2}{3} t$ $\therefore \frac{3g - 2v}{3g} = e^{-\frac{2t}{3}}$ $\therefore v = \frac{3g}{2} \left(1 - e^{-\frac{2t}{3}} \right)$	Correct integral $\rightarrow 1 \text{ mk}$ Correct integration $\rightarrow \frac{1}{2} \text{ mk}$ Substitution & simplify $\rightarrow \frac{1}{2} \text{ mk}$ Taking e to both sides $\rightarrow 1 \text{ mk}$	Variation to integral can be: $t = \int \frac{3dv}{3g - 2v}$ When $t = 0, v = 0$ $\rightarrow c = -\frac{3}{2} \ln(3g)$

1 (b) (ii)	as $t \rightarrow \infty$ $v \rightarrow \frac{3g}{2}$	Correct answer → 1 mk	
1 (b) (iii)	when $v = \frac{g}{2}$ $\therefore \frac{1}{3} = 1 - e^{-\frac{2t}{3}}$ $\therefore -\frac{2}{3}t = \ln\left(\frac{2}{3}\right)$ $\therefore t = -\frac{3}{2}\ln\left(\frac{2}{3}\right)$ or $t = \frac{3(\ln 3 - \ln 2)}{2}$	Correct substitution & simplification → 1 mk Taking logs of both sides & simplification → 1 mk	
1 (c) (i)	$P = \frac{QC}{C + e^{-kQt}}$ $\therefore \frac{dP(t)}{dt} = \frac{Q^2 C k e^{-kQt}}{(C + e^{-kQt})^2}$ $= \frac{kQC}{C + e^{-kQt}} \times \frac{Qe^{-kQt}}{C + e^{-kQt}}$ $= kP(Q - P)$ As $(Q - P) = Q - \frac{QC}{C + e^{-kQt}}$ $= \frac{QC + Qe^{-kQt} - QC}{C + e^{-kQt}}$ $= \frac{Qe^{-kQt}}{C + e^{-kQt}}$	Correct differential → 1 mk Simplification → 1 mk Showing $(Q - P) = \frac{Qe^{-kQt}}{C + e^{-kQt}} \rightarrow 1 \text{ mk}$	
1 (c) (ii)	as $t \rightarrow \infty$, $P \rightarrow Q$ as $e^{-kQt} \rightarrow 0$	Correct answer with some explanation → 1 mk	
1 (c) (iii)	as $t \rightarrow \infty$, $\frac{dP(t)}{dt} \rightarrow 0$ as $P \rightarrow Q$	Correct answer with some explanation → 1 mk	

Question 2		
(a) (i) $m\ddot{x} = m(-g) - mkv$ (upwards) $\therefore \ddot{x} = -g - kv \text{ i.e. } \frac{dv}{dt} = -g - kv$ $\therefore \frac{1}{k} \int_u^0 \frac{k dv}{-(g + kv)} = \int_0^T dt$ $\therefore -\frac{1}{k} [\ln g + kv]_u^0 = T$ $\therefore -\ln \left \frac{g}{g + ku} \right = kT$ $\therefore \ln \left \frac{g + ku}{g} \right = kT \Rightarrow kT = \ln \left 1 + \frac{ku}{g} \right $	Correct equation → 1 mk Correct integral → 1 mk Correct integration → 1 mk Correct simplification → 1 mk	
2 (a) (ii) Highest point reached is when $v = 0$ $\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -g - kv \Rightarrow v \frac{dv}{dx} = -g - kv$ $\therefore -\int_u^0 \frac{v dv}{g + kv} = \int_0^h dx$ $\therefore -\int_u^0 \frac{1}{k} - \frac{g dv}{k(g + kv)} = \int_0^h dx$ $\therefore -\int_u^0 1 - \frac{g dv}{(g + kv)} = kh$ $\therefore -\left[v - \frac{g}{k} \ln(g + kv) \right]_u^0 = kh$ $\therefore -\left[-u + \frac{g}{k} \ln \left(\frac{g + ku}{g} \right) \right] = hk$ $\therefore hk = u - gT \text{ as } kT = \ln \left 1 + \frac{ku}{g} \right \text{ from part(i)}$	$v \frac{dv}{dh} = -g - kv \rightarrow 1 \text{ mk}$ Correct integral → 1 mk $\frac{v dv}{g + kv} = \frac{1}{k} - \frac{g dv}{k(g + kv)} \rightarrow 1 \text{ mk}$ Correct integration → 1 mk Correct simplification & connecting answer from (i) → 1 mk	

2 (b) $xy = 4$ Eqn. of normal is $y = p^2x - 2p^3 + \frac{2}{p}$ (i) $\therefore$ coordinates of Q are $\left(2p - \frac{2}{p^3}, 0\right)$	Correct coordinates → 1 mk	
(ii) $M = \left[\left(\frac{2p + 2p - \frac{2}{p^3}}{2} \right), \left(\frac{\frac{2}{p}}{2} \right) \right]$ $M = \left[\left(2p - \frac{1}{p^3} \right), \frac{1}{p} \right]; p \neq 0$	Correct midpoint formula → 1 mk Correct simplification & restriction for p → 1 mk	Alternatively can award 1 mk each for x and y coordinate of midpoint.
(iii) $x = 2p - \frac{1}{p^3}; y = \frac{1}{p}$ $\therefore p = \frac{1}{y}; y \neq 0$ $\therefore x = \frac{2}{y} - y^3$ is the locus of $M; y \neq 0$	→ 1 mk → 1 mk equation → 1 mk restriction $y \neq 0$	
Question 3		
(a) (i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow y = \frac{b}{a} \sqrt{a^2 - x^2}$ $\therefore \text{Area} = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx$ $= \frac{4b}{a} \times \frac{1}{4} \pi a^2$ since $\int_0^a \sqrt{a^2 - x^2} dx$ is a quadrant of a circle, centre O radius a units. $\therefore \text{Area} = \pi ab$ sq. units.	→ 1 mk For $\frac{1}{4} \pi a^2$ → 1 mk → 1 mk explanation of using $\frac{1}{4} \pi a^2$	

3 (a) (ii)



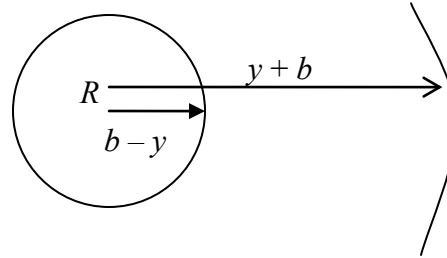
A slice taken through the ellipse perpendicular to the x – axis is the annulus with inner radius $(b - y)$ and outer radius $(b + y)$.

$$\therefore \text{Area of cross-section of slice} \\ = \pi[(b + y)^2 - (b - y)^2] = 4\pi by$$

$$\therefore \text{Volume of slice } \Delta V = 4\pi by \Delta x$$

$\therefore$ Volume of solid

$$\begin{aligned} V &= 4\pi b \int_{-a}^a y dx \\ &= 8\pi b \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx \\ &= \frac{8\pi b^2}{a} \times \frac{1}{4} \pi a^2 \text{ from part (i)} \\ &= 2\pi^2 ab^2 \text{ cubic units.} \end{aligned}$$

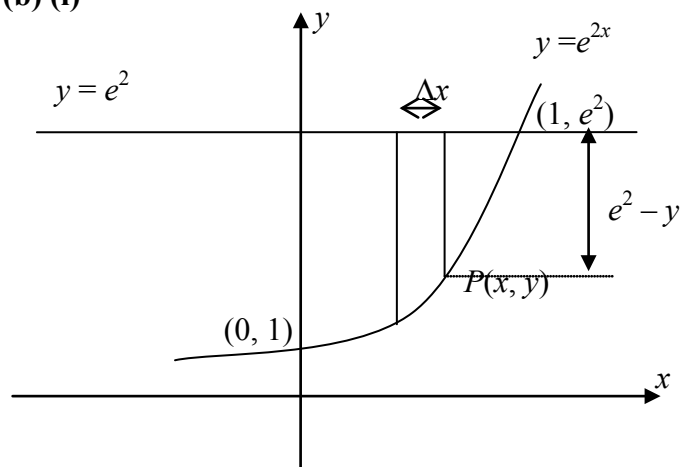


Showing area of cross section is $4\pi ay \rightarrow 1 \text{ mk}$

Correct integral $\rightarrow 1 \text{ mk}$

Correct Answer $\rightarrow 1 \text{ mk}$

3 (b) (i)



Area of rectangular slice; $A(x) = 2\pi x(e^2 - e^{2x})$

→ 1 mk

$$\therefore \Delta V = 2\pi x(e^2 - e^{2x})\Delta x$$

$$\therefore \text{Vol} = \lim_{\Delta x \rightarrow 0} \sum_0^1 2\pi x(e^2 - e^{2x})\Delta x$$

→ 1 mk

3 (b) (ii) $\therefore \text{Vol} = \int_0^1 2\pi x(e^2 - e^{2x})dx$

$$= \left[2\pi e^2 \cdot \frac{1}{2} x^2 \right]_0^1 - \pi \int_0^1 2x e^{2x} dx$$

→ 1 mk

$$= \pi e^2 - \pi [x e^{2x}]_0^1 + \pi \int_0^1 e^{2x} dx$$

→ 1 mk

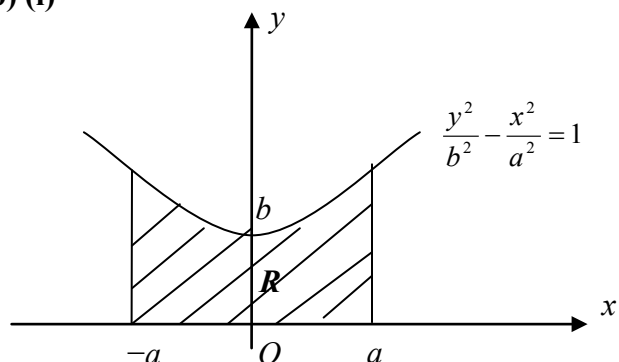
$$= \pi e^2 - \pi e^2 + \frac{1}{2} \pi [e^{2x}]_0^1 = \frac{1}{2} \pi (e^2 - 1)$$

→ 1 mk

3 (c) Substituting $y^2 = 4ax$ into $xy = c^2$ we get $y^3 = 4ac^2 = 2a^3$ as $2c^2 = a^2$. Let P be the point of intersection where $\therefore y = a\sqrt[3]{2}$ and $x = \frac{a(\sqrt[3]{4})}{4}$ By differentiating $xy = c^2$ we get $\frac{dy}{dx} = -\frac{y}{x}$ $\therefore$ the gradient of the hyperbola $xy = c^2$ at P is $m = \frac{-a(\sqrt[3]{2})}{\left(\frac{a}{4}\right)(\sqrt[3]{4})} = \frac{-4}{\sqrt[3]{2}} \rightarrow \boxed{A}$ By differentiating $y^2 = 4ax$ we get $\frac{dy}{dx} = \frac{2a}{y}$ $\therefore$ gradient of $y^2 = 4ax$ is given by $M = \frac{2a}{a(\sqrt[3]{2})} = \frac{2}{\sqrt[3]{2}} \rightarrow \boxed{B}$ $\therefore A \div B \rightarrow m = -2M$ as required.	$\rightarrow 1\text{mk}$ $\rightarrow \frac{1}{2} \text{ mk}$ $\rightarrow 1 \text{ mk}$ $\rightarrow \frac{1}{2} \text{ mk}$ $\rightarrow 1 \text{ mk}$	
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4(a) (i) $\ddot{x} = -\frac{k}{x^2}$ $\therefore \frac{d}{dx}\left(\frac{1}{2}v^2\right) = -\frac{k}{x^2}$ $\therefore \frac{1}{2}v^2 = -\int \frac{k}{x^2} dx = \frac{k}{x} + c$ Now $v = 0$ when $x = a \therefore c = -\frac{k}{a}$ $\therefore v^2 = 2k\left(\frac{1}{x} - \frac{1}{a}\right)$ Now $0 < x < a$ but motion is moving towards the origin for $t > 0$. $\therefore v = -\sqrt{2k\left(\frac{1}{x} - \frac{1}{a}\right)}$ For $x = \frac{1}{2}a$, $v = -\sqrt{\frac{2k}{a}}$	$\rightarrow \frac{1}{2}$ mk $\rightarrow \frac{1}{2}$ mk $\rightarrow 1$ mk Correct answer $\rightarrow 1$ mk	
4 (a) (ii) $\therefore \frac{1}{v} = \frac{dt}{dx} = -\frac{1}{\sqrt{2k\left(\frac{a-x}{ax}\right)}} = -\frac{1}{\sqrt{\frac{2k}{a}\left(\frac{a-x}{x}\right)}}$ $= -\sqrt{\frac{a}{2k}} \cdot \sqrt{\frac{x}{a-x}} = \sqrt{\frac{a}{2k}} \cdot \int_{\frac{a}{2}}^a \sqrt{\frac{x}{a-x}}$ $= -\sqrt{\frac{a}{2k}} \left[\sqrt{x(a-x)} + \frac{1}{2}a \sin^{-1}\left(\frac{a-2x}{a}\right) \right]_{\frac{a}{2}}^a$ $= -\sqrt{\frac{a}{2k}} \left[\frac{1}{2}a \sin^{-1}(-1) - \frac{a}{2} - \frac{1}{2}a \sin(0) \right]$ $\therefore t = \frac{(\pi + 2)a^{\frac{3}{2}}}{4\sqrt{2k}}$	Correct $\frac{1}{v}$ equation $\rightarrow 1$ mk $\sqrt{\frac{a}{2k}} \cdot \int_{\frac{a}{2}}^a \sqrt{\frac{x}{a-x}} \rightarrow 1$ mk Correct substitution $\rightarrow 1$mk	

4 (b) (i)



$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ can be written as $y = \pm \frac{b}{a} \sqrt{a^2 + x^2}$

$\therefore$ Area of region R is given by

$$\text{Area} = \frac{2b}{a} \int_0^a \sqrt{a^2 + x^2} dx$$

Let $x = a \tan \theta$. $\therefore$ at $x = a$, $\tan \theta = 1 \therefore \theta = \frac{\pi}{4}$

At $x = 0$, $\tan \theta = 0$ so $\theta = 0$. Also $dx = a \sec^2 \theta d\theta$

$$\therefore \text{Area} = A = \frac{2b}{a} \int_0^{\frac{\pi}{4}} \sqrt{a^2 + a^2 \tan^2 \theta} \cdot \sec^2 \theta d\theta$$

→ 1 mk

$$= 2ab \int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 \theta} \cdot \sec^2 \theta d\theta$$

$$= 2ab \int_0^{\frac{\pi}{4}} \sec^3 \theta d\theta \text{ where } I = \int \sec^3 \theta d\theta$$

→ 1 mk

4 (b) (i) Continued

$$\text{Now } I = \int \sec \theta (\sec^2 \theta) d\theta = \int \sec \theta (1 + \tan^2 \theta) d\theta$$

$$\begin{aligned} \therefore I &= \int \sec \theta d\theta + \int (\sec \theta \tan \theta) \tan \theta d\theta \\ &= \int \frac{\sec \theta (\sec \theta + \tan \theta)}{\sec \theta + \tan \theta} d\theta + \int \tan \theta \cdot \frac{d}{d\theta} (\sec \theta) d\theta \\ &= \ln (\sec \theta + \tan \theta) + \tan \theta \sec \theta - I \\ \therefore I &= \frac{1}{2} \ln (\sec \theta + \tan \theta) + \frac{1}{2} \tan \theta \sec \theta \end{aligned}$$

$$\text{So Area} = ab [\ln (\sec \theta + \tan \theta) + \tan \theta \sec \theta]_0^{\frac{\pi}{4}}$$

$$= ab [\sqrt{2} + \ln (1 + \sqrt{2})]$$

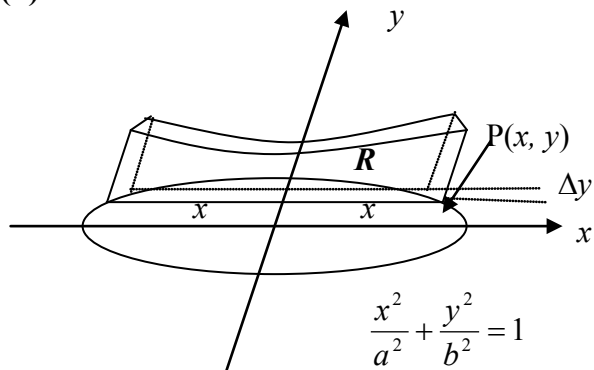
$$\text{Since } L = 2a \text{ area generated is } A = \frac{Lb}{2} [\sqrt{2} + \ln (1 + \sqrt{2})] \text{ unit}^2$$

→ 1 mk

→ 1mk

→ 1mk

4 (b) (ii)



Consider the cross section at $P(x, y)$ on the ellipse of thickness Δy (See diagram). The area of this cross section from (i) is A square units.

Note $L = 2a = 2x$ so $a = x$. $\therefore \text{Area} = A = xb \left[\sqrt{2} + \ln(1 + \sqrt{2}) \right]$ But $x = \frac{a}{b} \sqrt{b^2 - y^2}$ and let $K = \left[\sqrt{2} + \ln(1 + \sqrt{2}) \right]$ $\therefore A(y) = \frac{LK}{2} \sqrt{b^2 - y^2}$ Now volume of slice, $\Delta V = A(y) \Delta y$ $\therefore \Delta V = \frac{LK}{2} \sqrt{b^2 - y^2} \Delta y$ $\therefore$ volume of sum of slices, $V = \frac{LK}{2} \lim_{\Delta y \rightarrow 0} \sum_{-b}^b \sqrt{b^2 - y^2} \Delta y$ $= \frac{LK}{2} \int_{-b}^b \sqrt{b^2 - y^2} dy$ $= \frac{LK}{2} \cdot \frac{1}{2} \pi b^2$ (Note: this integral gives area of semi circle radius b) $\therefore$ Volume of $S = \frac{\pi L b^2}{4} \left[\sqrt{2} + \ln(1 + \sqrt{2}) \right]$ in terms of L and b or $= \frac{\pi a b^2}{2} \left[\sqrt{2} + \ln(1 + \sqrt{2}) \right]$ in terms of a and b.	$A(y) \rightarrow 1 \text{ mk}$ $\rightarrow 1 \text{ mk}$ $\rightarrow 1 \text{ mk}$ $\rightarrow 1 \text{ mk}$	
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~ End of Test ~